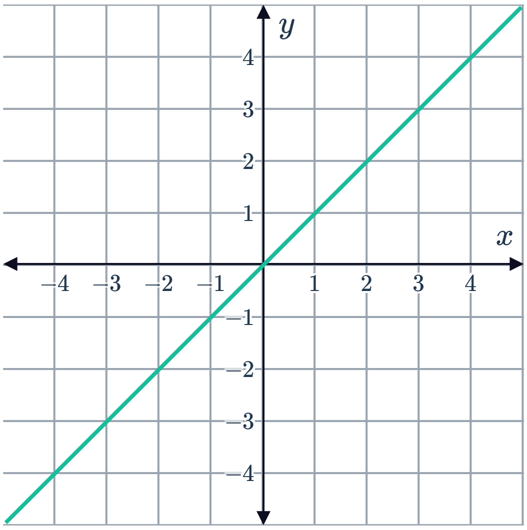
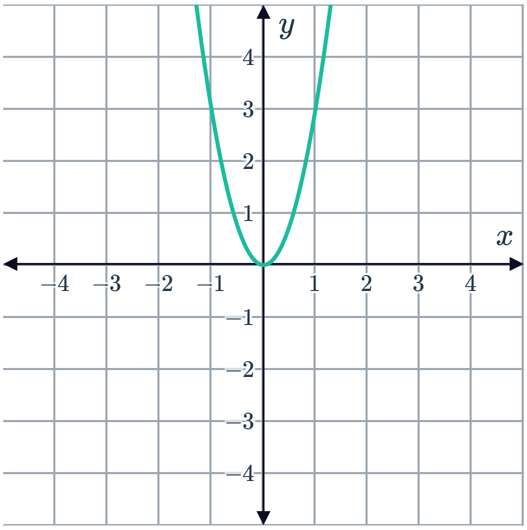
Power functions

1 Do the following graphed functions have an even or odd power?

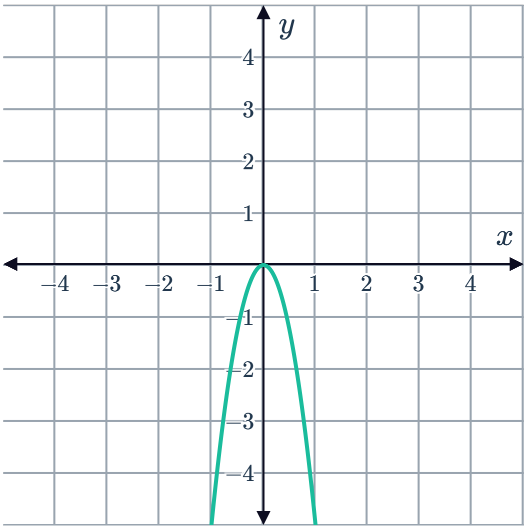
a



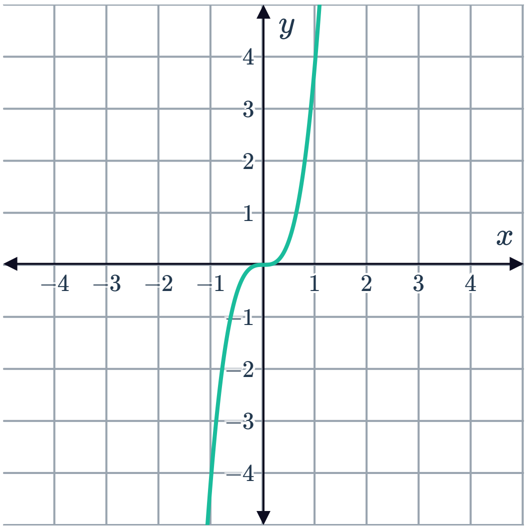
b



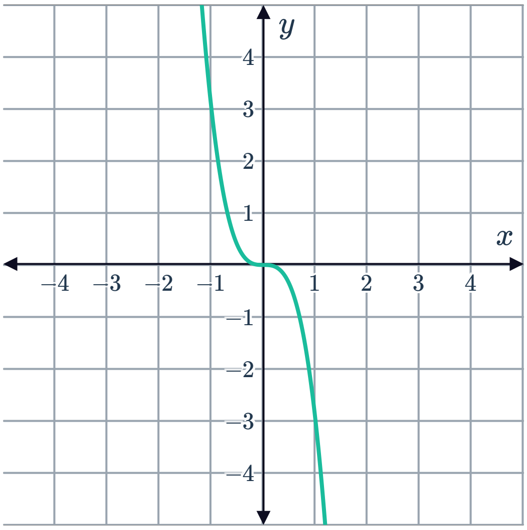
c



d



e



2 Consider the function $y = x^2$.

a Complete the following table of values:

x	-3	-2	-1	0	1	2	3
y							

b Using the points in the table, plot the curve on a cartesian plane.

c Are the y-values ever negative?

d Write down the equation of the axis of symmetry.

e What is the minimum y-value?

f For every y-value greater than 0, how many corresponding x values are there?

3 Consider the function $f(x) = -x^2$.

a Does the graph rise or fall to the right?

b Does the graph rise or fall to the left?

4 Consider the functions $f(x) = -x^4$ and $g(x) = -x^6$.

a Graph $f(x) = -x^4$ and $g(x) = -x^6$ on the same set of axes.

b Which of the above functions has the narrowest graph?

5 Consider the functions $f(x) = x^3$ and $g(x) = x^5$.

a Graph $f(x) = x^3$ and $g(x) = x^5$.

b How would the graph of $y = x^7$ differ to the graph of $f(x) = x^3$ and $g(x) = x^5$?

6 Consider the function $y = x^7$.

a As x approaches infinity, what happens to the corresponding y-values?

b As x approaches negative infinity, what happens to the corresponding y-values?

c Sketch the general shape of $y = x^7$.

d Sketch the general shape of $y = -x^7$.